





Parity



Wigner

Motivation:

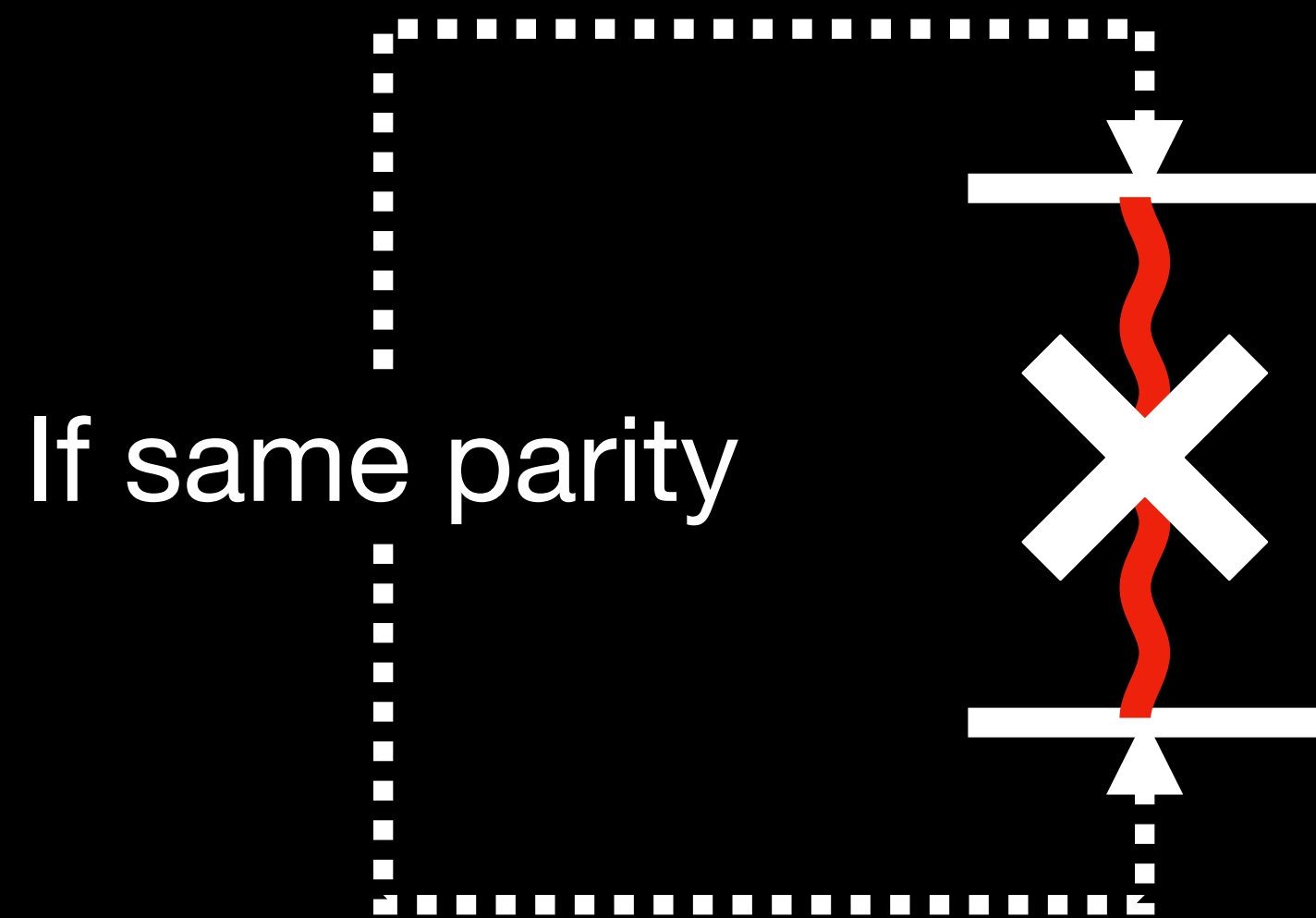
- Selection rule in atomic spectra (Laporte, Russel)
- Wigner showed (1927) that energy spectra have two symmetries:

$$\psi(-\mathbf{r}) = \psi(\mathbf{r})$$

Positive Parity: $P = +1$

$$\psi(-\mathbf{r}) = -\psi(\mathbf{r})$$

Negative Parity: $P = -1$



Photon has $P = -1$

Parity was assumed to be conserved

$$\tau - \theta$$

puzzle